

## 2024 AMC 10A Problem 17 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 17. Great practice for AMC 10, AMC 12, AIME, and other math contests

### 2024 AMC 10A Problem 17

**Problem:**

Two teams are in a best-two-out-of-three playoff: the teams will play at most 3 games, and the winner of the playoff is the first team to win 2 games. The first game is played on Team A's home field, and the remaining games are played on Team B's home field. Team A has a  $\frac{2}{3}$  chance of winning at home, and its probability of winning when playing away from home is  $p$ . Outcomes of the games are independent. The probability that Team A wins the playoff is  $\frac{1}{2}$ . Then  $p$  can be written in the form  $\frac{1}{2}(m - \sqrt{n})$ , where  $m$  and  $n$  are positive integers. What is  $m + n$ ?

**Answer Choices:**

- A. 10
- B. 11
- C. 12
- D. 13
- E. 14

**Solution:**

There are three ways for Team A to win the playoff: win the first two games; win the first game, lose the second game, and win the third game; or lose the first game and win the second and third games. The probability that it wins in one of these ways is

$$\frac{2}{3} \cdot p + \frac{2}{3} \cdot (1 - p) \cdot p + \frac{1}{3} \cdot p^2 = -\frac{1}{3}p^2 + \frac{4}{3}p$$

Setting this equal to  $\frac{1}{2}$  and simplifying gives  $2p^2 - 8p + 3 = 0$ , and the Quadratic Formula gives solutions  $\frac{1}{2}(4 \pm \sqrt{10})$ . Choosing the plus sign gives a nonsensical value of  $p$  because it is greater than 1, so the required probability is  $\frac{1}{2}(4 - \sqrt{10}) \approx 0.42$ . The requested sum is  $4 + 10 = (E)\boxed{14}$ .

*The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗*

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